$$k = -\frac{\ln(0.1)}{\ln(2)} = -\frac{-2,3025}{0,6931} = 3,32$$

$$m = -n \cdot \frac{\ln(p)}{\ln(2)^2} = 4,792 \cdot n$$

4.11 Improved Variants

Bloom filters have been around for almost 50 years, so it's natural that many variations and improvements have been proposed in the meantime.

Let's examine some of them, with a focus on those that improve accuracy.

4.11.1 Bloomier Filter

As we hinted at, Bloom filters corresponds to faster, lighter versions of *HashSets*, since they can only store if a key is present/absent.

The leaner counterpart of *HashTables* has been introduced only recently: *Bloomier filters*[22] allow associating values to keys. When the key/value pair has been actually stored in the Bloomier filter, then the value returned is always correct. There are still *false positives*, i.e. keys that were not actually stored in the data structure but for which a value is returned.

4.11.2 Combining Bloom Filters